

SYSTEM DESIGN DOCUMENT

(SDD)

Blinkit Sales Performance Dashboard — Power BI Technical Design

Document Type	System / Solution Design Document
Related Artefact	blinkit_dashboard.pbix
Platform	Microsoft Power BI Desktop (Fabric report definition format)
Version	1.0
Status	Draft — Reverse-engineered from delivered .pbix file

1. Introduction & Purpose

This System Design Document (SDD) describes the technical design of the **Blinkit Sales Performance Dashboard**, a Power BI (.pbix) report. It covers the data model, calculation layer (DAX measures), report/page architecture, visual inventory, and interactivity design, as reverse-engineered from the report's internal definition files. This document should be read alongside the Process Design Document (PDD), which covers the business objectives and process context.

Important note on method: Power BI stores its data model (tables, relationships, and DAX formulas) inside a compressed, engine-level binary section of the .pbix file. This section is not human-readable through static file inspection. This document's data-model and measure descriptions were therefore derived from the fields and measures actually referenced across the report's visuals (which are stored in readable JSON), combined with standard Power BI/DAX conventions. Exact DAX formula syntax should be confirmed by opening the file in Power BI Desktop.

2. Architecture Overview

The solution follows a standard single-file Power BI architecture:

- **Data Source Layer:** Data was imported (Import mode, not DirectQuery/live) into the Power BI file; the file's internal settings confirm no relationships were auto-created on load, consistent with a single flat table structure.
- **Data Model Layer (VertiPaq engine):** Holds one primary data table ("Table1"), one measures table ("DAX MEASURES") used purely to organise calculations, and one small helper table ("Metrics") used to drive a dynamic selector slicer.
- **Semantic/Calculation Layer:** DAX measures defined on top of Table1 provide the KPIs used throughout the report.
- **Presentation Layer:** A single Power BI report page (1920×1080, "Page 1") containing KPI cards, charts, slicers, a pivot table, and two bookmark-linked navigation buttons.

3. Data Model Design

3.1 Tables identified in the model

Table	Type	Purpose
Table1	Primary data table (imported)	Holds outlet and item-level sales records — item attributes, outlet attributes, sales and rating figures.
DAX MEASURES	Measures table (calculation group)	A dedicated, empty table used only to organise and host the report's DAX measures, per common Power BI best practice.
Metrics	Disconnected helper table	A small standalone table (single column "Metrics") used to drive a slicer that lets the user switch which metric is emphasised in the report.

3.2 Relationships

The file's Settings metadata explicitly records **zero auto-created relationships**, confirming a single-table (denormalised) model rather than a star schema. "DAX MEASURES" and "Metrics" are used as disconnected tables (a common pattern for measure organisation and for building dynamic field/metric selectors via slicers).

3.3 Columns referenced by the report

Table	Column	Used In
Table1	Item Type	Pivot table rows, funnel chart category
Table1	Item Fat Content	Clustered bar chart category
Table1	Outlet Establishment Year	Pivot table rows (summed), area chart category
Table1	Outlet Size	Donut chart category, funnel category, slicer, pivot table rows
Table1	Outlet Type	Pivot table columns
Table1	Outlet Location Type	Slicer
Metrics	Metrics	Slicer (dynamic metric selector)

These are the columns confirmed via direct field references inside the report's visuals. The source table may contain additional columns (e.g. Item Identifier, Item MRP, Item Visibility, Outlet Identifier) that exist in the model but are not directly placed on any visual, and therefore could not be enumerated from the report definition alone.

4. DAX Measures

The following measures are defined in the "DAX MEASURES" table and consumed across the KPI cards and charts. Exact DAX syntax is stored in the compressed data model and is inferred here from standard naming/aggregation conventions:

Measure	Likely Definition (inferred)	Used In
Total sales	SUM of the item/outlet sales column	KPI card, funnel (Item Type), clustered bar chart
Average sales	AVERAGE of the item/outlet sales column	KPI card, donut chart, clustered bar chart, area chart, pivot table
Total outlets	DISTINCTCOUNT of the outlet identifier column	KPI card
Average Rating	AVERAGE of the outlet/item rating column	KPI card, donut chart, funnel (Outlet Size), area chart, pivot table

Recommendation: confirm the exact DAX expressions by opening the .pbix file in Power BI Desktop (Model view → select each measure → view its formula bar) and update this section with verbatim formulas for full technical accuracy.

5. Report & Page Design

- **Canvas:** Single page ("Page 1"), 1920 × 1080 px, "Fit to Page" display mode.
- **Theme:** Dark background using a theme accent colour; Blinkit brand wordmark rendered as a text box ("BLINK" in black + "IT" in green, Candara Bold 80pt).
- **Layout zones:** Branding/title top-left, four KPI cards top-right, slicers on the left rail, charts filling the middle/lower canvas, and a detailed pivot table overlay reachable via navigation.

6. Visual Inventory

#	Visual Type	Field(s)	Purpose
1	Text box	—	Blinkit brand title/wordmark
2	Card	Average sales	KPI headline
3	Card	Total sales	KPI headline
4	Card	Total outlets	KPI headline
5	Card	Average Rating	KPI headline
6	Slicer	Metrics (helper table)	Dynamic metric selector
7	Slicer	Outlet Location Type	Filter by location tier
8	Slicer	Outlet Size	Filter by outlet size
9	Donut chart	Outlet Size (cat.) / Avg Rating, Avg sales (values)	Rating & sales split by outlet size
10	Funnel chart	Item Type (cat.) / Total sales (value)	Sales ranked by item category

#	Visual Type	Field(s)	Purpose
11	Funnel chart	Outlet Size (cat.) / Avg Rating (value)	Rating ranked by outlet size
12	Clustered bar chart	Item Fat Content (cat.) / Avg sales, Total sales (values)	Sales by fat-content classification
13	Area chart	Outlet Establishment Year (cat.) / Avg Rating, Avg sales (values)	Trend over outlet establishment year
14	Pivot table (matrix)	Rows: Item Type, Outlet Size, Establishment Year; Columns: Outlet Type; Values: Avg sales, Avg Rating	Detailed cross-tab drill-down view
15	Shape (pill) ×2	Bookmark actions	"HOME" and "TABLE" navigation buttons
16	Shape (rectangle)	—	Background panel
17	Visual group	Wraps pivot table	Groups the table view for show/hide via bookmarks

7. Interactivity & Navigation Design

The report uses **Power BI Bookmarks** to implement a two-view toggle experience on a single page:

Bookmark	Behaviour
HOME	Restores the default chart-based summary view (KPI cards, donut, funnels, bar and area charts visible).
TABLE	Reveals the detailed pivot table view (visual group), for granular drill-down analysis.

Two pill-shaped shape visuals are configured with bookmark navigation actions, acting as "HOME" and "TABLE" buttons that the user clicks to switch views. This avoids the need for a second report page and keeps the entire experience on one canvas.

In addition, three slicers (Metrics, Outlet Location Type, Outlet Size) provide standard cross-filtering across all visuals on the page.

8. Technical Specifications

Item	Value
File format	.pbix (Power BI report, Fabric-style report definition/PBIR structure)
Authoring tool	Power BI Desktop
Report definition version	Report page schema v2.1.0 / visual container schema v2.12.0
Data connectivity mode	Import (no DirectQuery/live connection detected)
Relationships	None auto-created ("AutoCreatedRelationships": [] in Settings) — single flat table model
Report pages	1 ("Page 1"), 1920×1080
Bookmarks	2 ("HOME", "TABLE")
Created from	Cloud (per file Metadata), Power BI release 2026.08

9. Performance Considerations

- A single flat table with no relationships keeps query complexity low, which should give fast visual rendering for moderate data volumes.
- As the dataset grows, consider migrating to a star schema (separate outlet and item dimension tables linked to a sales fact table) to reduce data duplication and improve model efficiency.
- The "Metrics" disconnected table pattern is lightweight but should be reviewed to confirm it doesn't require row-by-row iteration that could slow down interactions on larger datasets.

10. Testing & Validation Approach

- Validate each KPI card total against a manual aggregation (e.g. SUM/AVERAGE) of the source data in Excel or SQL.
- Verify slicer interactions correctly cross-filter all visuals on the page.

- Verify the HOME and TABLE bookmark buttons correctly show/hide the intended visuals without leaving orphaned elements visible.
- Cross-check the pivot table's cross-tabulated figures against the summary charts for consistency.

11. Deployment & Maintenance

- Current deployment: standalone .pbix file, shareable directly or via Power BI Service workspace publish.
- For ongoing use, establish a scheduled refresh (if migrated to Power BI Service with a connected data source) to keep the dataset current.
- Maintain this SDD alongside any future changes to the data model, measures, or report layout.

Prepared based on structural analysis of the supplied .pbix file's report definition layer. The compressed internal data model could not be parsed for verbatim DAX formulas; verify exact expressions directly in Power BI Desktop.